# 机器学习
# 第 2 章 模型评估与选择

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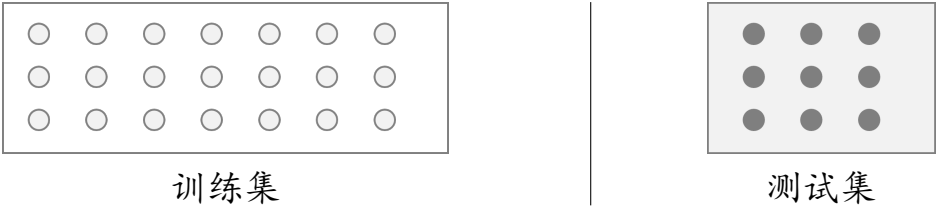
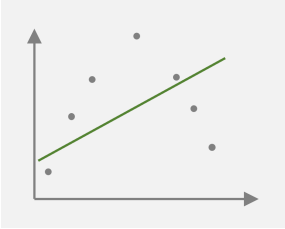
# 1. 机器学习流程

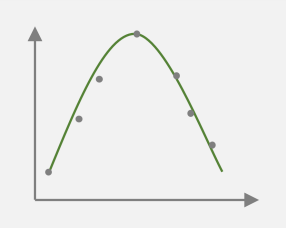


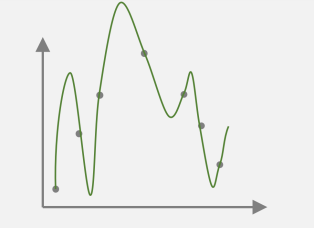

训练模型 $\longrightarrow$ 测试模型

训练模型 $\longrightarrow$ 模型选择与调参 $\longrightarrow$ 测试模型

# 2. 模型选择与超参数选择



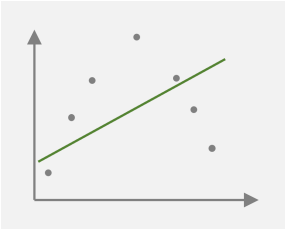


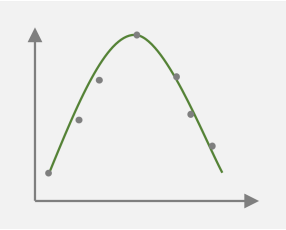


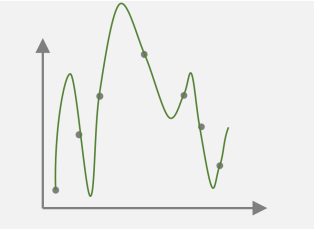
$f = \sum_{i=1}^{k} w_i x^i + w_{k+1}$, $k$ 为超参数

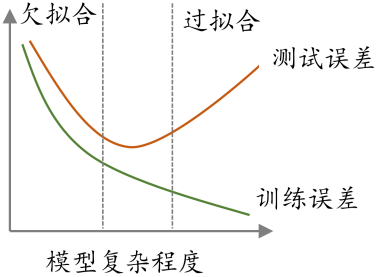
**选在训练集上误差最小的模型？**

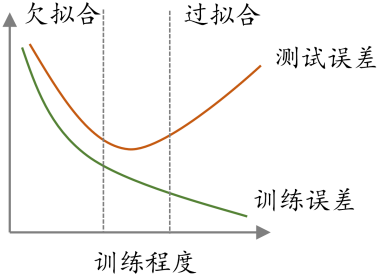
# 2. 模型选择与超参数选择:欠拟合与过拟合



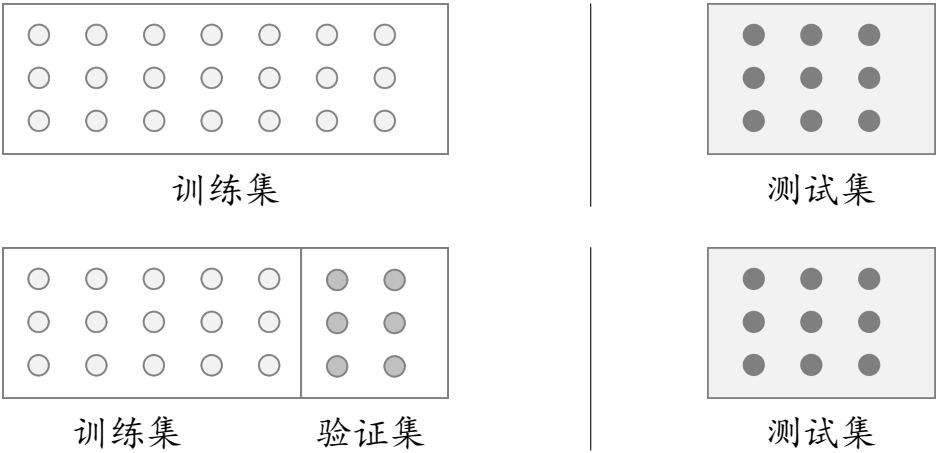






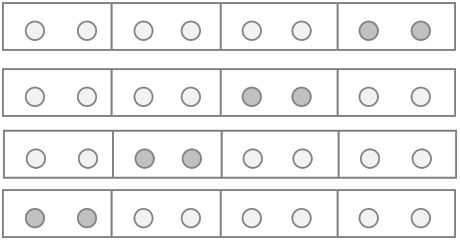
# 2. 模型选择与超参数选择



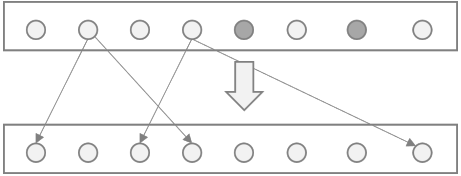
# 2. 模型选择与超参数选择: 划分验证集的方法

留出法

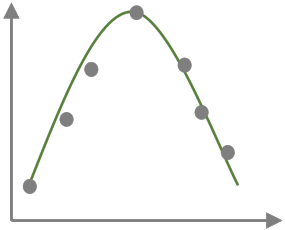
交叉验证



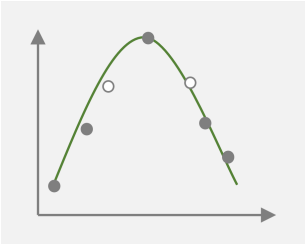
自助法



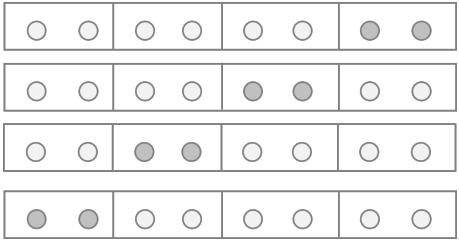
# 2. 模型选择与超参数选择: 留出法



留出法 ⟶



# 2. 模型选择与超参数选择: 交叉验证法



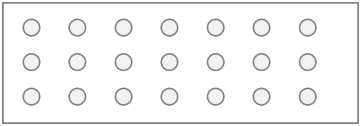
| 模型 A | 模型 B |
|---|---|
| $e_{A1}$ | $e_{B1}$ |
| $e_{A2}$ | $e_{B2}$ |
| $e_{A3}$ | $e_{B3}$ |
| $e_{A4}$ | $e_{B4}$ |

比较 4 折交叉验证误差平均值

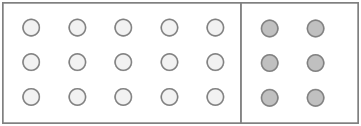
$$\frac{1}{4}\sum_{i=1}^{4} e_{Ai} < \frac{1}{4}\sum_{i=1}^{4} e_{Bi}$$

# 2. 模型选择与超参数选择: 完整训练流程

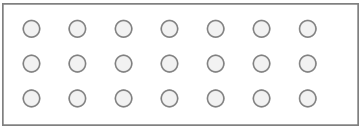
训练集



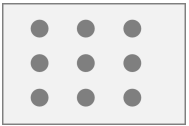
模型选择与调参



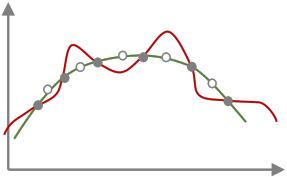
完整训练



交付使用

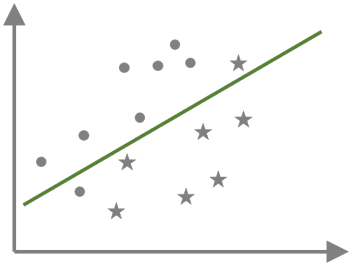


# 3. 性能评估: 回归问题



- 平方和误差 SSE $= \sum_{i=1}^{m}(f_i - y_i)^2$
- 均方误差 MSE $= \frac{1}{m}\sum_{i=1}^{m}(f_i - y_i)^2$
- 均方根误差 RMSE $= \sqrt{\frac{1}{m}\sum_{i=1}^{m}(f_i - y_i)^2}$
- 平均绝对误差 MAE $= \frac{1}{m}\sum_{i=1}^{m}|f_i - y_i|$

# 3. 性能评估: 分类问题



- 准确率 $acc = \frac{1}{m}\sum_{i=1}^{m}\mathbb{I}(f_i = y_i)$
- 错误率 $err = \frac{1}{m}\sum_{i=1}^{m}\mathbb{I}(f_i \neq y_i)$
- $acc + err = 1$
- 二分类与多分类都适用

# 3. 性能评估: 二分类问题

$acc = 0.5$

$acc = 0.9$

**对于类别不均衡问题，错误模型有高准确率，不合理！**

## 3. 性能评估: 二分类问题的混淆矩阵

| 真实 \ 预测 | 正例 | 反例 |
|---|---|---|
| 正例 | $TP$ | $FN$ |
| 反例 | $FP$ | $TN$ |

- 查准率：$P = \dfrac{TP}{TP + FP}$
- 查全率：$R = \dfrac{TP}{TP + FN}$
- $F_1$ 值：$F_1 = \dfrac{2PR}{P + R}$

# 3. 性能评估: 二分类问题的混淆矩阵

| 真实 \ 预测 | 正例 | 反例 |
|---|---|---|
| 正例 | $TP = 1$ | $FN = 3$ |
| 反例 | $FP = 0$ | $TN = 6$ |

- 查准率：$P = \dfrac{TP}{TP + FP} = \dfrac{1}{1+0} = 100\%$
- 查全率：$R = \dfrac{TP}{TP + FN} = \dfrac{1}{1+3} = 25\%$
- $F_1$ 值：$F_1 = \dfrac{2PR}{P+R} = \dfrac{2 \times 1 \times 0.25}{1 + 0.25} = 40\%$

# 3. 性能评估: 作业 1

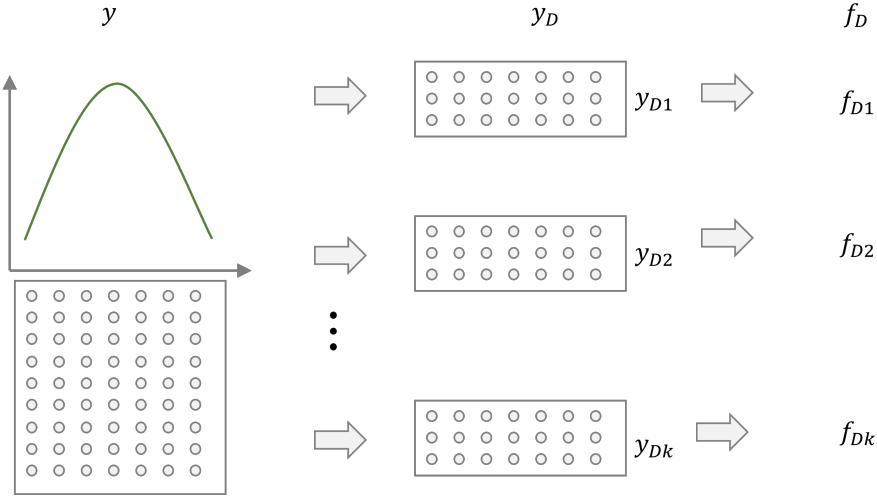
| 序号 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| 标签 | 阴性 | 阴性 | 阴性 | 阴性 | **阳性** | 阴性 | 阴性 | 阴性 | 阴性 | 阴性 |
| 预测 | 阴性 | 阴性 | 阴性 | 阴性 | **阳性** | 阴性 | **阳性** | 阴性 | 阴性 | 阴性 |

| 序号 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
|---|---|---|---|---|---|---|---|---|---|---|
| 标签 | 阴性 | 阴性 | 阴性 | 阴性 | 阴性 | 阴性 | 阴性 | 阴性 | 阴性 | **阳性** |
| 预测 | 阴性 | 阴性 | 阴性 | 阴性 | 阴性 | 阴性 | 阴性 | 阴性 | **阳性** | **阳性** |

| 真实 \ 预测 | 正例 | 反例 |
|---|---|---|
| 正例 | $TP = 2$ | $FN = 0$ |
| 反例 | $FP = 2$ | $TN = 16$ |

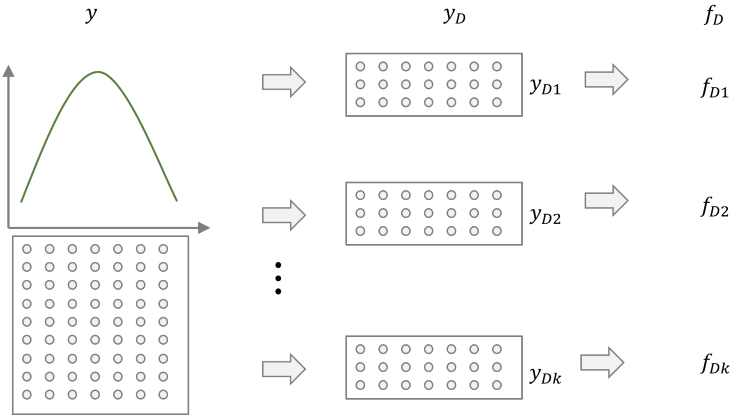
- 查准率：$P = \dfrac{2}{2+2} = 50\%$
- 查全率：$R = \dfrac{2}{2+0} = 100\%$
- $F_1$ 值：$F_1 = \dfrac{2 \times 0.5 \times 1}{0.5 + 1} = 67\%$
- 准确率：$acc = \dfrac{18}{20} = 90\%$
- 错误率：$err = \dfrac{2}{20} = 10\%$

# 4. 偏差与方差




$$y_D(\boldsymbol{x}) = y(\boldsymbol{x}) + \epsilon, \epsilon \sim N(0, \sigma^2)$$

# 4. 偏差与方差




**期望泛化误差：**

$$\mathbb{E}_D\left[(f_D - y_D)^2\right]$$

**偏差：**

$$(y - \mathbb{E}_D[f])^2$$

**方差：**

$$\mathbb{E}_D\left[(f_D - \mathbb{E}_D[f_D])^2\right]$$

# 4. 偏差与方差: 基础知识

## (1) 随机变量 $X$，期望 $\mathbb{E}[X]$

- $\mathbb{E}[X+Y] = \mathbb{E}[X] + \mathbb{E}[Y]$
- $\mathbb{E}[aX] = a\mathbb{E}[X]$
- $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$ 当 $X$ 与 $Y$ 独立

## (2) 方差 $Var[X]$

$$
\begin{aligned}
Var[X] &= \mathbb{E}\left[(X - \mathbb{E}[X])^2\right] \\
&= \mathbb{E}\left[X^2 - 2X\mathbb{E}[X] + \mathbb{E}[X]^2\right] \\
&= \mathbb{E}[X^2] - 2\mathbb{E}[X]^2 + \mathbb{E}[X]^2 \\
&= \mathbb{E}[X^2] - \mathbb{E}[X]^2
\end{aligned}
$$

$$
\mathbb{E}\left[(X - \mathbb{E}[X])^2\right] = \mathbb{E}[X^2] - \mathbb{E}[X]^2
$$

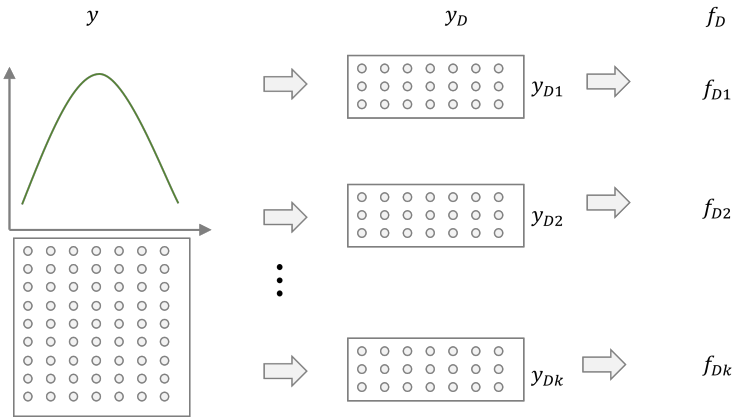
## 4. 偏差与方差

**期望泛化误差：**

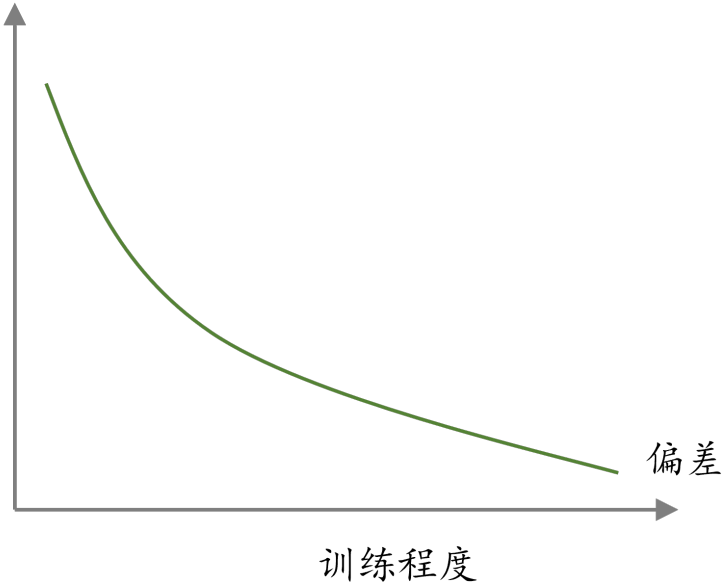
$$
\begin{aligned}
&\mathbb{E}_D\left[(f_D-y_D)^2\right]\\
=&\mathbb{E}_D\left[f_D^2-2f_Dy_D+y_D^2\right]\\
=&\mathbb{E}_D[f_D^2]-2\mathbb{E}_D[f_Dy_D]+\mathbb{E}_D[y_D^2]\\
=&\left(\mathbb{E}_D\left[(f_D-\mathbb{E}_D[f_D])^2\right]+\mathbb{E}_D[f_D]^2\right)-2\mathbb{E}_D[f_D(y+\epsilon)]+\mathbb{E}_D[(y_D+\epsilon)^2]\\
=&\left(\mathbb{E}_D\left[(f_D-\mathbb{E}_D[f_D])^2\right]+\mathbb{E}_D[f_D]^2\right)-2\mathbb{E}_D[f_Dy+f_D\epsilon]+\mathbb{E}_D[y^2+2y\epsilon+\epsilon^2\\
=&\left(\mathbb{E}_D\left[(f_D-\mathbb{E}_D[f_D])^2\right]+\mathbb{E}_D[f_D]^2\right)-2y\mathbb{E}_D[f_D]+y^2+\mathbb{E}_D[\epsilon^2]\\
=&\mathbb{E}_D\left[(f_D-\mathbb{E}_D[f_D])^2\right]+(\mathbb{E}_D[f_D]-y)^2+\mathbb{E}_D[\epsilon^2]\\
=&\mathbb{E}_D\left[(f_D-\mathbb{E}_D[f_D])^2\right]+(\mathbb{E}_D[f_D]-y)^2+\sigma^2
\end{aligned}
$$

# 4. 偏差与方差

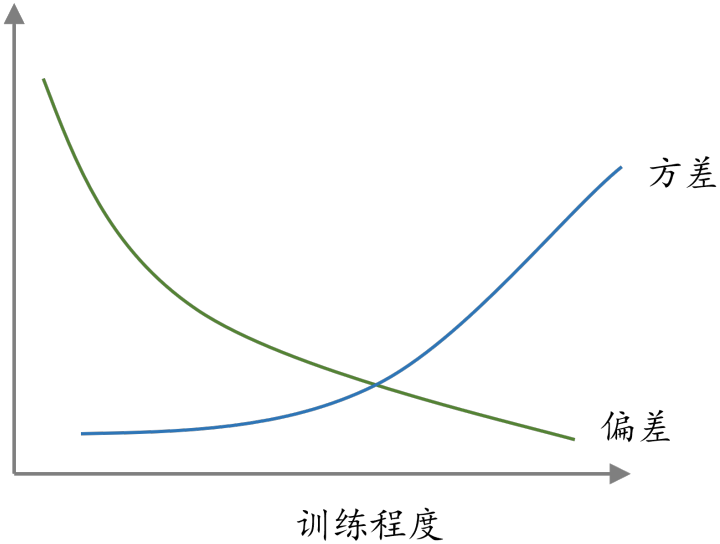
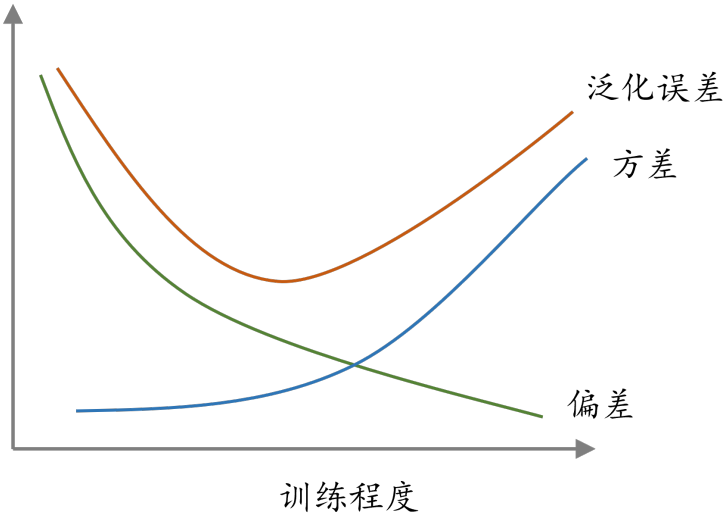
$$\mathbb{E}_D\left[(f_D - y_D)^2\right] = (y - \mathbb{E}_D[f_D])^2 + Var[f_D] + \sigma^2$$



# 4. 偏差与方差



# 4. 偏差与方差



# 4. 偏差与方差